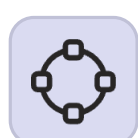




Inventory and Resource Sharing



Shared Consumables Policy

Lab personnel will prioritize sharing consumables, reagents, and chemicals using the Quartzy inventory management system to avoid duplicate purchases.



Shared Equipment Use

Lab personnel will utilize shared equipment. There are many items that are in our lab's inventory that are designated for general lab use.



Surplus Resource Management

The lab will repair, donate, donate, or send surplus equipment, consumables, and chemicals to refurbishment organizations or other end-users who can benefit from them. (MERCI, (MERCI, ROSE, ReUSE)

Inventory Check Before Orders

Ensure that existing inventory in Quartzy is checked before placing new orders to prevent excessive or redundant purchasing.

Order Consolidation Policy

Orders will be consolidated into larger shipments whenever possible to reduce the environmental impact of transportation.

Lab Water Use and Quality

We are committed to promoting responsible water use and ensuring the appropriate application of various water qualities for laboratory tasks. To achieve this, the lab provides comprehensive training sessions focused on effective water-saving strategies and the proper use of water types, including tap, deionized, distilled, and reverse osmosis-purified water. All personnel are required to select water quality that is fit for purpose, avoiding unnecessary over-purification to prevent waste and reduce operational costs.



Waste Management



Safe Waste Disposal Guidelines

Lab personnel will adhere to guidelines for the safe disposal of liquid waste, separating aqueous, non-aqueous, and halogenated (chlorinated) waste when possible.



Minimizing Water-Based Vacuums

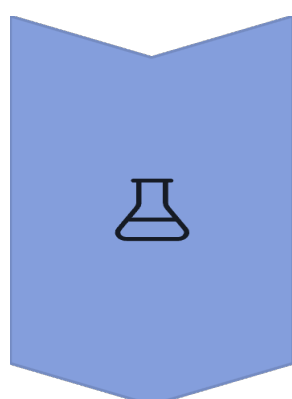
Labs will not use water-vacuum aspirators or other water-based methods to create a vacuum.

Energy Efficiency

Implementing energy-efficient practices is essential for reducing the laboratory's environmental footprint and operational costs.

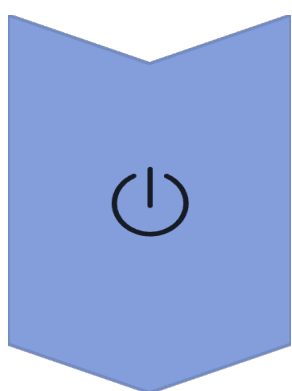
Additional Energy Policies

- Biosafety cabinets must be turned off completely when not in use for extended periods.
- Cloud First Data Storage Policy: The lab encourages storing data and files in the cloud rather than on local networks or computers to reduce energy usage and improve accessibility. Each UVA One Drive account permits up to 5TB of storage.



Cold Trap Usage

A cold trap must be used in line before vacuum pumps to prevent volatiles from entering the pump.



Vacuum Pump Use

Vacuum pumps must be turned off when not in use to save energy.



Incubators

Lab personnel are required to turn incubators completely off when they are not in use to conserve energy.



Biosafety Cabinets

Biosafety cabinets must be turned off completely when not in use for extended periods.



Environmental Practices

Sample and Reagent Storage Policy

All laboratory personnel are required to store samples and reagents at the minimum appropriate temperature specified by manufacturer guidelines, research protocols, or institutional standard operating procedures. Storing materials at unnecessarily low temperatures can lead to potential degradation, increased storage costs, and inefficient use of laboratory resources. Personnel must carefully review specific storage requirements for each type of sample or reagent, utilizing refrigeration or freezer units that precisely match the recommended temperature range. Periodic audits will be conducted to verify compliance with these storage guidelines and to prevent unnecessary overcooling of laboratory materials.

Ambient Conditions Preference

Lab personnel will evaluate and, when possible, switch protocols, procedures, or reactions to ambient conditions instead of using elevated or reduced temperatures and pressures.

Virtual Meetings and Green Travel

Personnel are encouraged to use virtual meetings, alternative local transportation, and trains instead of short-haul flights for conferences and external meetings whenever feasible.

Fume Hood and Safety Practices



Fume Hood Sash Use

Personnel must not work in the fume hood with the sash pulled above the sash stop (typically 18" or 45 cm) to ensure safety and energy efficiency.

No Storage in Fume Hoods

Chemicals, reagents, consumables, excess equipment, or containers must not be stored in fume hoods.

By adhering to these policies, the lab aims to promote sustainable practices, optimize resource use, and maintain a safe working environment for all personnel.